Asset Performance:Ene 04 Air permeability of the fabric

Contributes to 40 credits available for asset energy rating

No Minimum Standard

Applicability: All projects

Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and of installed buildings services

Question

Enter the result of the building pressure/air leakage test if known ($m^3/h.m^2@50Pa$)

Credits	Enter pressure/ air leakage test result
-	$m^3/h.m^2@50Pa$

If a pressure/air leakage test has not been conducted; has a thermographic survey been undertaken?

Credits	Answer	Select a single answer option
-	A.	Yes
-	B.	No

Assessment criteria

Criterion	Assessment criteria
1.	Testing must be conducted by a relevant competent person.

Methodology

Air leakage testing standard

The appropriate standard for air leakage testing is: ISO 9972:2015 Thermal performance of buildings - Determination of air permeability of buildings - Fan pressurization method

Air leakage testing

Air leakage testing results must be from testing that has, at least, been carried out after construction of the building or when structural changes have been made to the building.

Thermographic survey standard

The appropriate standard for a Thermographic survey is ISO 6781:1983 Thermal insulation -- Qualitative detection of thermal irregularities in building envelopes -- Infrared method

Thermographic survey

The thermographic survey should cover 100% of the treated spaces, unless it is a large complex building and ensure that all elements of the building fabric that enclose an internal heated or conditioned (treated) zone of the building will be tested. This includes internal walls separating treated and untreated zones.

In the case of large and complex buildings, e.g. airports, large hospitals and high-rise buildings, it may be impractical for the thermographic survey and air tightness testing to cover 100% of the building. Where a complete thermographic survey is deemed impractical by a Class/Category II thermographic surveyor, the guidance in air tightness standard ISO 9972:2015 should be followed regarding the extent of the survey and testing.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Copy of results from building pressure and/or air leakage test.
All	Copy of thermographic survey results.
All	Confirmation of competence levels for persons performing testing.

Definitions

Air leakage test:

A test which quantifies the air permeability rate of the building envelope. The more airtight the building fabric is the lower the air permeability result will be. To maximise energy efficiency, it is advised that the air permeability result is as low as reasonably practicable.

Relevant competent person:

For Air leakage testing:

An individual achieving all of the following can be considered to be a relevant competent person:

a) Holds a recognised qualification in airtightness testing and measurement.

b) Has relevant experience in air pressure testing and has carried out testing on at least ten large non-residential buildings within the last five years. They should also have a recognised qualification in airtightness testing and measurement.

Their expertise should be broad enough to cover all required technical aspects guaranteeing that the data collected during the test is appropriate and the results reflect the assessed airtightness performance of the building. It can be someone operating as sole trader or employed by public or private enterprise bodies.

For thermographic surveys:

Professionals holding a valid Category II certificate in thermography, as defined by ISO 18436-7:2014 or Class II certificate in infrared thermography as defined by ISO 6781-3:2015.